

D656

Leveraging AI and Technology in Design

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A design agency with offices in Denver and Austin demos an image-tagging tool trained on 2 million product photos. A client asks whether the tool “understands” furniture the way a person does and could also draft its legal contracts. What is the most accurate answer?

- A. It follows a rulebook of furniture definitions written by engineers.
- B. Yes; with 2 million photos it has general knowledge of furniture and law.
- C. AI can never tag images accurately.
- D. It learned visual patterns from its training data for tagging; it does not understand furniture and is not built for contracts.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

A UX writer at a health-insurance startup that recently rebranded uses a chatbot to draft onboarding copy. The draft states that members get “free dental cleanings twice a year,” which sounds right. What should the writer do before publishing?

- A. Publish it, since the chatbot is trained on insurance content.
- B. Rewrite the draft to match the new brand voice, then publish.
- C. Verify the benefit against the actual plan documents and correct it if needed.
- D. Never use AI for any writing.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3. [TRANSFER — reversed stem / different variable]

A research summary made by AI cites three journal articles. A reviewer cannot find two of them anywhere. What is the most likely explanation?

- A. The articles are behind paywalls, so they must exist.
- B. The tool generated plausible but nonexistent citations.
- C. AI never makes up sources.
- D. The tool was biased against those journals.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A photo app's skin-smoothing filter, built by a team of 14 engineers, works well on light skin but blurs features on darker skin. Its training set was 85% images of light-skinned faces. What is the most likely cause?

- A. The team was too small.
- B. The training data under-represented darker skin, so the model learned skewed patterns.
- C. Darker skin is too hard for any filter to handle.
- D. Algorithms are neutral math, so it must be a camera problem.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

A team wants to prevent bias before launching an AI résumé screener. Which step addresses the root cause?

- A. Audit the training data for skew and test outcomes across demographic groups before launch.
- B. Add a nicer interface for recruiters.
- C. Remove the job title field from the form.
- D. Launch and fix bias only if someone complains.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A property manager with 1,200 units wants AI to approve or deny rental applications automatically to cut review time from three days to one hour. The vendor's dashboard won a design award. What oversight is appropriate?

- A. Use AI to flag and rank, but have a person review denials and be accountable for final decisions.
- B. Fully automate it since it is faster.
- C. Trust it because the dashboard won an award.
- D. Automate it only for the first 1,000 units.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7. [TRANSFER — reversed stem / different variable]

Which task can most reasonably run with little human review?

- A.** Any task, as long as the AI is accurate 90% of the time.
- B.** Auto-generating alt-text drafts that a designer spot-checks before release.
- C.** Deciding which employees are laid off.
- D.** Approving medical claims.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A bank's support chat uses an AI agent named "Sam" with a friendly photo chosen after 30 rounds of A/B testing. Customers assume Sam is a person. What should the design team do?

- A.** Remove the chatbot entirely.
- B.** Keep it as is, since the photo tested well.
- C.** Replace the photo with a different face.
- D.** Clearly label Sam as an AI assistant and offer a path to a human.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A researcher has 25 interview transcripts containing patient names and diagnoses from a hospital client. She wants a free public AI tool to summarize them before a Friday deadline. What should she do?

- A.** Upload them, since the tool is free and widely used.
- B.** Upload only 10 of the 25 to reduce risk.
- C.** Upload them because the deadline is close.
- D.** Not upload them unless the data is de-identified and the client's agreement and the tool's terms allow it.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

An agency generates hero images for a paid national campaign using a prompt that names a famous living illustrator's style. The images took four minutes to make. What should the team check before using them?

- A. Only the image resolution.
- B. Whether four minutes is fast enough to bill.
- C. The tool's commercial license, the client's policy on AI assets, and the risk of imitating a named artist.
- D. Nothing; AI output is always free to use.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A design team of five spends a combined 20 hours a week resizing assets for social channels. An AI tool, which has a dark-mode interface, cuts that work by 75%. How many hours per week does the team save?

- A. 15 hours
- B. 3.75 hours
- C. 5 hours
- D. 20 hours

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12. [TRANSFER — reversed stem / different variable]

A task took 12 hours a week before an AI tool and takes 3 hours after. By what percent did the tool reduce the time?

- A. 25%
- B. 300%
- C. 75%
- D. 9%

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer D Concept 1

A design agency ~~with offices in Denver and Austin~~ demos an image-tagging tool trained on 2 million product photos. A client asks whether the tool “understands” furniture the way a person does and could also draft its legal contracts. What is the most accurate answer?

Target word: what narrow ML can do **Decides:** Narrow AI learned patterns for one task.

A	Confusion set	Describes rule-based systems, not machine learning.
B	Wrong stem number	Training volume does not create general intelligence.
C	Absolute language	Image tagging is a strength of narrow ML.
D	Correct	Narrow ML is task-specific pattern recognition.

2. Answer C Concept 2

A UX writer at a health-insurance startup ~~that recently rebranded~~ uses a chatbot to draft onboarding copy. The draft states that members get “free dental cleanings twice a year,” which sounds right. What should the writer do before publishing?

Target word: handling generative output **Decides:** Generated claims must be verified.

A	Inverted operation	Treats plausibility as proof.
B	Camouflage	Voice is not the risk; accuracy is.
C	Correct	Generative AI can state false details confidently.
D	Absolute language	AI drafts are fine with human review.

3. Answer B Concept 2 · transfer

A research summary made by AI cites three journal articles. A reviewer cannot find two of them anywhere. What is the most likely explanation?

Target word: reverse from symptom to cause **Decides:** Hallucinated citations.

A	Inverted operation	Assumes existence without evidence.
B	Correct	Hallucinated references are a known failure mode.
C	Absolute language	It can and does.
D	Confusion set	Bias is a different failure; this is fabrication.

4. Answer B Concept 3

A photo app’s skin-smoothing filter, ~~built by a team of 14 engineers~~, works well on light skin but blurs features on darker skin. Its training set was 85% images of light-skinned faces. What is the most likely cause?

Target word: source of bias **Decides:** Skewed training data.

A	Camouflage	Team size does not explain uneven results; the data does.
B	Correct	Models mirror their data.
C	Absolute language	Balanced data fixes much of this.
D	Inverted operation	Denies data-driven bias.

5. Answer A Concept 3 · transfer

A team wants to prevent bias before launching an AI résumé screener. Which step addresses the root cause?

Target word: prevention rather than diagnosis **Decides:** Audit and balance training data; test by group.

A	Correct	Targets the data and measures results by group.
B	Adjacent category	Usability, not fairness.
C	Dropped element	A small tweak that ignores the data source.
D	Inverted operation	Reactive, not preventive.

6. Answer A Concept 4

A property manager with 1,200 units wants AI to approve or deny rental applications automatically to cut review time from three days to one hour. ~~The vendor's dashboard won a design award.~~ What oversight is appropriate?

Target word: oversight for high-stakes decisions **Decides:** Housing access is high impact; a human must review.

A	Correct	High-stakes decisions need human review and a way to appeal.
B	Inverted operation	Speed does not outweigh impact on people.
C	Camouflage	Interface awards say nothing about fairness.
D	Wrong stem number	The unit count does not change the stakes.

7. Answer B Concept 4 · transfer

Which task can most reasonably run with little human review?

Target word: low-stakes automation **Decides:** Low impact on people and easily reversed.

A	Absolute language	Accuracy alone does not remove the need for oversight on high-impact tasks.
B	Correct	Low stakes, reviewed, and easy to fix.
C	Inverted operation	The highest-stakes kind of decision.
D	Adjacent category	Affects health and money; needs oversight.

8. Answer D Concept 5

A bank's support chat uses an AI agent named "Sam" with a friendly photo ~~chosen after 30 rounds of A/B testing.~~ Customers assume Sam is a person. What should the design team do?

Target word: disclosure **Decides:** Users should know they are talking to AI.

A	Absolute language	Disclosure fixes the issue without dropping the feature.
B	Camouflage	Test performance does not justify deception.
C	Dropped element	Still hides AI use.
D	Correct	Transparency builds trust and respects user choice.

9. Answer D Concept 6

A researcher has 25 interview transcripts containing patient names and diagnoses from a hospital client. She wants a free public AI tool to summarize them ~~before a Friday deadline~~. What should she do?

Target word: privacy in AI use **Decides:** Personal health data plus an unapproved third-party tool.

A	Inverted operation	Popularity does not make it safe.
B	Wrong stem number	Any identifiable record is a problem.
C	Camouflage	Deadlines do not override privacy duties.
D	Correct	Personal health data needs consent and approved handling.

10. Answer C Concept 7

An agency generates hero images for a paid national campaign using a prompt that names a famous living illustrator's style. ~~The images took four minutes to make~~. What should the team check before using them?

Target word: IP risk of generated assets **Decides:** Commercial use and imitation of a named artist.

A	Adjacent category	A quality check, not a rights check.
B	Camouflage	Speed has no bearing on rights.
C	Correct	Ownership and rights are uncertain and must be checked.
D	Absolute language	Licenses and rights vary.

11. Answer A Concept 8

A design team of five spends a combined 20 hours a week resizing assets for social channels. An AI tool, ~~which has a dark mode interface~~, cuts that work by 75%. How many hours per week does the team save?

Target word: hours saved **Decides:** 20×0.75 .

A	Correct	$20 \times 75\% = 15$ hours saved; 5 hours remain.
B	Wrong stem number	Divides by the team size of five.
C	Inverted operation	That is the time remaining, not saved.
D	Absolute language	Assumes AI removes all of the work.

12. Answer C Concept 8 • transfer

A task took 12 hours a week before an AI tool and takes 3 hours after. By what percent did the tool reduce the time?

Target word: solve for percent reduction **Decides:** $(12 - 3) / 12$.

A	Inverted operation	$3 / 12$ is the share remaining.
B	Confusion set	$(12 - 3)/3$ uses the new time as the base.
C	Correct	$9 / 12 = 75\%$.
D	Dropped element	Uses the hour difference and drops the base.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	3	
Wrong stem number	A figure copied from the prompt that is not the answer.	4	
Adjacent category	The right idea in the wrong bucket.	3	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	9	
Absolute language	Always, never, guarantee — where judgment applies.	8	
Confusion set	Every option is real; the test is exact role-matching.	3	
Camouflage	Answers a stem fact that was never the question.	6	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Machine learning learns from data	Machine learning systems find patterns in training data rather than following hand-written rules; today's AI is narrow, built for specific tasks.	Thinking current AI understands or reasons like a person across any task (general AI).	A claim that a tool "knows" or "understands" something beyond its training.
2	Generative output is probable, not verified	Generative AI produces plausible output that can be wrong (hallucination), so outputs need human checking before use.	Treating fluent, confident output as fact.	AI-written copy, citations, or statistics headed to publication.
3	Bias comes from data and design	Models reflect the patterns, gaps, and skew in their training data and objectives, so they can reproduce or amplify bias.	Believing algorithms are neutral because they are math.	Uneven results across groups that were under-represented in the data.
4	Human oversight scales with stakes	The higher the impact of a decision on people, the more a human must review, override, and be accountable for AI output.	Fully automating consequential decisions for speed.	AI output that affects access, money, safety, or reputation.
5	Transparency and disclosure	Users should know when they are interacting with AI or AI-generated content and, where it matters, why the system produced a result.	Hiding AI use to make a product feel more human or authoritative.	A chatbot or generated image presented without labeling.
6	Privacy and data handling	Do not put confidential, personal, or client data into AI tools unless the tool's data terms and the owner's consent allow it.	Pasting user research transcripts into a public tool because it is convenient.	Sensitive data plus a third-party AI service.
7	Ownership and licensing of generated assets	Check the tool's license, the training data rights, and the client's requirements before using generated assets commercially; imitation of a named living artist's style raises added risk.	Assuming anything an AI generates is free to use.	Generated imagery bound for a paid campaign or product.
8	AI augments productivity	AI yields the most value on repetitive, well-defined tasks, freeing people for judgment work; measure gains as time or cost saved against a baseline.	Expecting AI to replace design judgment, or claiming gains with no baseline.	Hours per task before and after a tool.

Course scope (public description): Leveraging AI and Technology in Design provides students with the current landscape of artificial intelligence (AI), how it has evolved, and its impact on design. Students will gain the skills necessary to use AI and other new technologies in ethical and innovative ways, lead teams in devising ethical guidelines for using AI in designing digital products and enhance organizational productivity using design technology.